

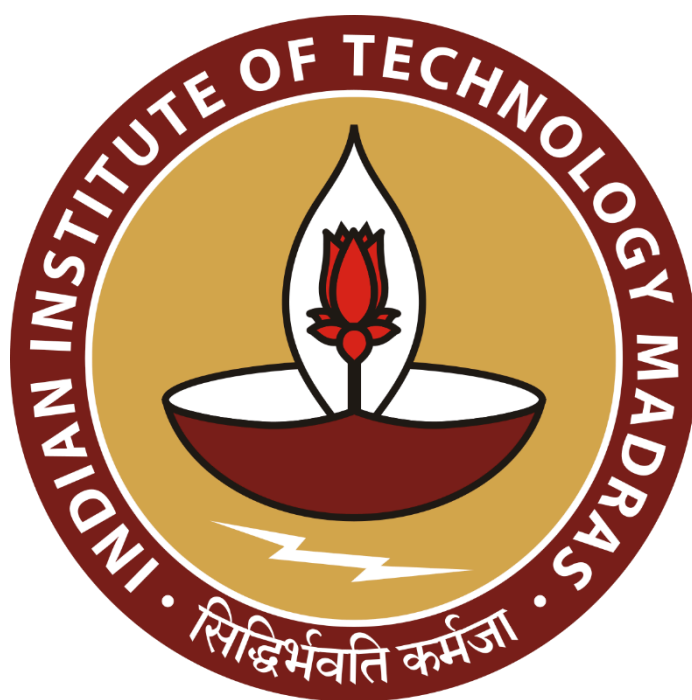
Data-Driven Optimization and Scalability of The Diabetes Care Medical and General Stores in Retail Pharmacy sector

A final submission report for the BDM capstone Project

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1. Executive Summary and Title

Project Title: Data-Driven Optimization and Scalability of The Diabetes Care Medical and General Stores in Retail Pharmacy sector.

Diabetes Care Medical and General Stores, a profitable pharmacy established in 2010, faces two critical barriers to growth. The firm struggles to scale and open a new branch by not leveraging available supplier credit, which ties up working capital. This is compounded by chronic inventory imbalances, where outdated systems cause frequent, costly overstocking and stockout situations.

This project analyzed 10 months of operational data (April 2024 - January 2025) to solve these problems. For scalability, a comprehensive financial risk assessment was conducted; this included a negotiation model to reclaim trapped capital, a 12-month cash flow forecast to model the new branch's risks, and a reinvestment capacity analysis of the main store. For the inventory problem, a statistical analysis was performed, including visualizing supply versus demand, building a new Safety Stock (SS) and Reorder Point (ROP) model, and quantifying the financial impact.

The analysis yielded significant, actionable financial insights. The firm can safely self-fund its expansion by unlocking ₹1,140,000 in trapped supplier capital (identified via a Decision Tree model) and reallocating ₹454,373 from the main branch's highly stable profits. The new inventory policy was also financially justified, with the analysis identifying over ₹54,558 in cash currently trapped in just the top 10 overstocked items, including ₹18,556 from METSMALL 500mg alone.

Based on these findings, this report recommends a two-part solution. First, to solve scalability, the owner must initiate "Partnership" negotiations with top suppliers and formally earmark the main branch profits to fully self-fund the new branch. Second, to solve the inventory problem, the pharmacy must immediately adopt the new SS & ROP model for all purchases and place a temporary freeze on the top 10 overstocked items. Implementing these recommendations will unlock significant capital and provide a clear, data-driven path to sustainable growth.

2. Detailed Explanation of Analysis Process/Method

2.1. Data Cleaning and Preprocessing:

- **Explanation:** The data preparation for this project was a two-stage process. The first stage involved a foundational cleanup of the 10 months of raw operational records, from April 2024 to January 2025. This data was sourced from two main master datasets: a purchase and GST dataset and a separate inventory and sales dataset. Both were organized with different sheets for each of the 10 months. This initial cleanup

was essential for reliability and included standardizing all monetary figures to Rupees, unifying inventory units to "strips" (1 strip = 10 tablets), and pruning irrelevant data to create a master dataset.

The second stage, which was critical for the advanced analyses in this final report, involved data transformation and aggregation. Instead of just using the raw logs, I had to create several new, model-ready tables. For the Supply vs. Demand analysis, I programmatically extracted Sale Quantity and Purchase Quantity for specific medicines across all 10 monthly sheets and aggregated them into a new time-series table. For the Safety Stock model, I built a comprehensive 20x10 matrix by pulling all sales data, which became the new basis for statistical calculations like STDEV and AVERAGE. Finally, for the Financial Impact analysis, I created another new table by merging Price, Opening Stock, and Closing Stock data from all 10 files with the results from the Safety Stock model.

- **Importance:** This two part process was vital. The initial cleanup ensured the data was trustworthy. The second stage data transformation was what truly enabled the advanced analysis. By building these specific, aggregated datasets, I was able to move beyond simple reporting and instead build complex, predictive models, like the Safety Stock formula or the Reinvestment Capacity chart, which are the core of this final report.

2.2. Explanation of Method/Analysis Used:

Starting with the **first problem statement**, regarding scaling up and capital utilization of the organization, below are the required analysis based on previous midterm report submission analysis:

- 2.2.1. **Negotiation Impact Model & Decision Tree Analysis:** This analysis provides a direct solution to the scalability problem by creating a strategic action plan. The midterm report identified significant trapped capital; this analysis models the risk versus reward of reclaiming it from the two highest impact suppliers, Solapur Pharma LLB and Baldawa Enterprises. It moves beyond simply identifying the problem to simulating the financial outcomes of specific negotiation strategies.

To build this model, I first quantified the Benefit as the Total Trapped capital from the previous analysis.. I then defined a Risk Cost by assuming a 5% price hike on the Total Purchase Amount from that supplier, modeling the potential backlash. Two strategies were then defined: a low conflict "Partnership" path with an 80% success probability and a high conflict "Aggressive" path with a 60% success probability.

Using these variables, I calculated the Expected Value (EV) for all four scenarios. The core mathematical formulas used are:

$$\text{Risk Cost} = \text{Total Purchase Amount} * 0.05(5\% \text{ assumed rate for price hike})$$

$$\text{Expected Value} = (\text{Benefit} * \text{Success Probability}) - (\text{Risk Cost} * \text{Failure Probability})$$

Justification: This analysis was justified because simply identifying trapped capital was not enough to solve the scalability problem. A practical action plan was needed, so this model was chosen to simulate the financial consequences of real world negotiation. It turns a vague goal into a data driven decision, proving which strategy is financially superior and giving the owner a clear path to unlock the capital needed for expansion.

2.2.2. New Branch 12-Month Cash Flow & Sensitivity Analysis: This analysis directly addresses the scalability problem by modeling the financial viability of the new branch after it is funded. While the midterm report identified the best funding scenario, this analysis projects the new branch's actual performance over its first 12 months. I created a 12 month cash flow model based on the key assumptions from the previous midterm analysis, including the ₹1,600,000 setup cost and the ₹750,000 target annual profit. To build the model, I had to first define the monthly operational costs, which were a Base Fixed OpEx (rent, salary) of ₹75,000 and a Base COGS Margin (Cost of Goods Sold) of 80%. This allowed for the calculation of the Base Monthly Sales needed to hit the profit target. The model then projects the Monthly Net Cash Flow and Cumulative Cash Flow to identify the break even month.

The second part was a sensitivity analysis to test risk, where I measured the impact of a 10% change in Sales and Fixed OpEx, and a 1% change in COGS margin, on the final annual profit.

Mathematical Equations: The core of the model was calculating the required sales. The Base Monthly Profit was first found:

$$\text{Base Monthly Profit} = \text{Target Annual Profit} / 12$$

Then, the Base Monthly Sales was calculated from the profit formula:

$$\text{Base Monthly Sales} = (\text{Base Monthly Profit} - \text{Base Fixed OpEx}) / (1 - \text{Base COGS})$$

Finally, the Monthly Net Cash Flow for the operational months was calculated as:

$$\text{Monthly Net CashFlow} = \text{Monthly Sales} - \text{COGS} - \text{Fixed OpEx}(\text{rent/salary})$$

Justification: This model was justified because proving the best funding source does not guarantee the new branch will be successful. This analysis moves to the depth of a dynamic operational question like answering, "What are the real financial risks of this expansion?" The sensitivity analysis was chosen to provide a clear, visual answer to this, highlighting for the owner which variable, Sales, has the most impact on profitability, thereby showing them what they must control for the expansion to succeed.

2.2.3. Main Branch Profitability & Reinvestment Capacity Analysis: This analysis was conducted to validate the financial stability of the primary recommendation for

Problem Statement 1. The midterm report's winning "Scenario 1" required using ₹454,373 of profit from the main branch to help fund the new store. This analysis proves that the existing branch is profitable enough to provide these funds without putting its own operations at risk.

I used the "PURCHASE DATA SUMMARY" data sheet for each of the ten months to accomplish this. I started by figuring out each month's actual monthly net profit in rupees. To demonstrate how the amount of money available for reinvestment increased over the course of the ten months, I then computed the Cumulative Net Profit month over month.

Mathematical Equations: Two straightforward but essential computations for every month served as the foundation for the analysis. The monthly net profit in rupees was first determined.

$$\text{Monthly Net Profit (₹)} = \text{Total Sales (₹)} * \text{Net Profit (\%)}$$

Second, the Cumulative Net Profit for any given month was calculated by adding the current month's profit to the previous month's total:

$$\text{Cumulative Net Profit}(i) = \text{Cumulative Net Profit}(i - 1) + \text{Monthly Net Profit}(i)$$

where $\text{Cumulative Net Profit}(i)$ is Cumulative Net Profit of i th Month.

Justification: If the main branch was not financially stable, it would be reckless to recommend using its profits, which is why this analysis was justified. This model was selected because it offered unambiguous, visual evidence of the main branch's soundness and ability to finance the expansion. By verifying that the main branch can securely supply its portion of the funding, it directly supports the scalability solution and makes the expansion plan as a whole safe and feasible.

For the second problem statement, focused on optimizing the pharmacy's inventory management and resolving the frequent stock imbalances, the following in-depth analyses were performed:

2.2.4. Monthly Supply vs. Demand Imbalance Analysis: This analysis serves as the visual starting point for solving Problem Statement 2. The midterm report identified inventory imbalances; this analysis visualizes the behavior of that imbalance over time for two specific, problematic medicines: LANTUS VIAL, which was identified as a low turnover, overstocked item, and STARVOG 0.2mg, which was a high turnover, stockout risk item.

To create this analysis, I aggregated data from all 10 monthly inventory sheets. For each of the two medicines, I created a new time series table plotting their Purchase Quantity (representing "Supply") and their Sale Quantity (representing "Demand") for each month from April 2024 to January 2025. This data was then visualized on two separate line charts, with "Supply" as one line and "Demand" as another, to provide a clear, month by month comparison of how stock levels became imbalanced.

Mathematical Equations: This method is a direct data visualization rather than a complex statistical model. The core of the analysis is the graphical representation of two functions over time (t):

$$\text{Supply}(t) = \text{Purchase Quantity (at month } t)$$

$$\text{Demand}(t) = \text{Sale Quantity (at month } t)$$

Justification: This analysis was justified because it provides a simple, intuitive, and powerful visual confirmation of the inventory problem. Instead of just relying on a single ITR (Inventory Turnover Ratio) number as calculated in the midterm analysis report is not enough, this method shows the owner the dynamic behavior that causes the problem. It was chosen to set a clear foundation for the next analysis, by proving why a simple guesswork approach to purchasing fails and why a statistical model is necessary.

2.2.5. Safety Stock (SS) & Reorder Point (ROP) Statistical Model: This analysis is the primary statistical solution to Problem Statement 2. It was designed to replace the pharmacy's current guesswork inventory method, which causes frequent stock imbalances, with a data driven and objective policy. To build this model, I used the 10 month sales data for all 20 medicines. First, I calculated the Average Daily Demand for each item, as well as the Standard Deviation of Daily Demand, which is a key measure of its sales unpredictability.

With these calculations, I established two key operational assumptions: a Supplier Lead Time of 3 days and a desired Service Level of 95 percent (which provides a standard Z Score of 1.65). These variables were then applied to standard industry formulas to calculate two critical numbers for each medicine: the Calculated Safety Stock, which is the buffer stock needed to prevent stockouts, and the Calculated Reorder Point, which is the stock level that triggers a new order.

Mathematical Equations: First, the daily demand statistics were calculated.

$$\text{Avg. Daily Demand} = \text{Total Sale Quantity} / \text{Total Days in Period}$$

Because sales data was monthly, the daily variability was found using Standard Deviation of Daily Demand. Next, the Calculated Safety Stock was found using its formula:

$$\text{Safety Stock} = Z \text{ Score} * (\text{StDev of Daily Demand}) * \text{SQRT}(\text{Lead Time})$$

Finally, the Calculated Reorder Point was determined:

$$\text{Reorder Point} = (\text{Avg. Daily Demand} * \text{Lead Time}) + \text{Safety Stock}$$

Justification: This model was chosen because the pharmacy's core inventory problem stems from a lack of any data driven reordering system. This analysis directly solves that problem by providing a specific, statistical, and proven method for managing every item. It moves beyond simply identifying the problem to providing a complete,

actionable solution, giving the owner the exact Safety Stock and Reorder Point numbers needed to create an efficient and stable inventory policy.

2.2.6. Financial Impact of Inventory Optimization Analysis: This final analysis connects Problem Statement 2 directly back to Problem Statement 1 by quantifying the exact financial savings, in Rupees, of implementing the new inventory model. To achieve this, I created a new summary table.

First, I brought in the *Optimized_Inventory_Strips* for each item, which is the Calculated Safety Stock figure from our Analysis 5 model. These two different inventory levels were then multiplied by the *Avg_Price* to get a "Current Capital Trapped" value and an "Optimized Capital" value.

Mathematical Equations: The analysis is based on three key financial calculations. First, the Current Capital Trapped in Rupees was calculated as:

$$\text{Current Capital Trapped (₹)} = \text{Current_Avg_Inventory_Strips} * \text{Avg_Price}$$

Second, the Optimized Capital needed on the shelf, based on the new model, was found using:

$$\text{Optimized Capital (₹)} = \text{Optimized_Inventory_Strips} * \text{Avg_Price}$$

Finally, the Potential Savings was calculated by finding the difference between these two values:

$$\text{Potential Savings (₹)} = \text{Current Capital Trapped (₹)} - \text{Optimized Capital (₹)}$$

Justification: This analysis was justified because it provides the crucial factor for the entire inventory investigation, moving beyond just a statistical model to show the tangible, financial benefit of adopting the new policy. By calculating the exact Rupee amount of cash trapped in overstocked items, it provides a powerful financial incentive for the owner. It directly links the inventory solution back to the primary scalability problem by identifying another source of capital that can be unlocked for expansion.

3. Results and Findings

For analysis/methods of Problem Statement 1:

3.1. Negotiation Impact Model & Decision Tree Analysis:

This analysis calculated the Expected Value (EV) for two different negotiation strategies with the top two suppliers. For Solapur Pharma LLB, the "Partnership" strategy yielded an EV of ₹275,921, while the "Aggressive" strategy yielded a much lower EV of ₹188,680. A similar result was found for Baldawa Enterprises, where the "Partnership" strategy's EV of ₹235,478 was significantly higher than the "Aggressive" path's EV of ₹158,609.

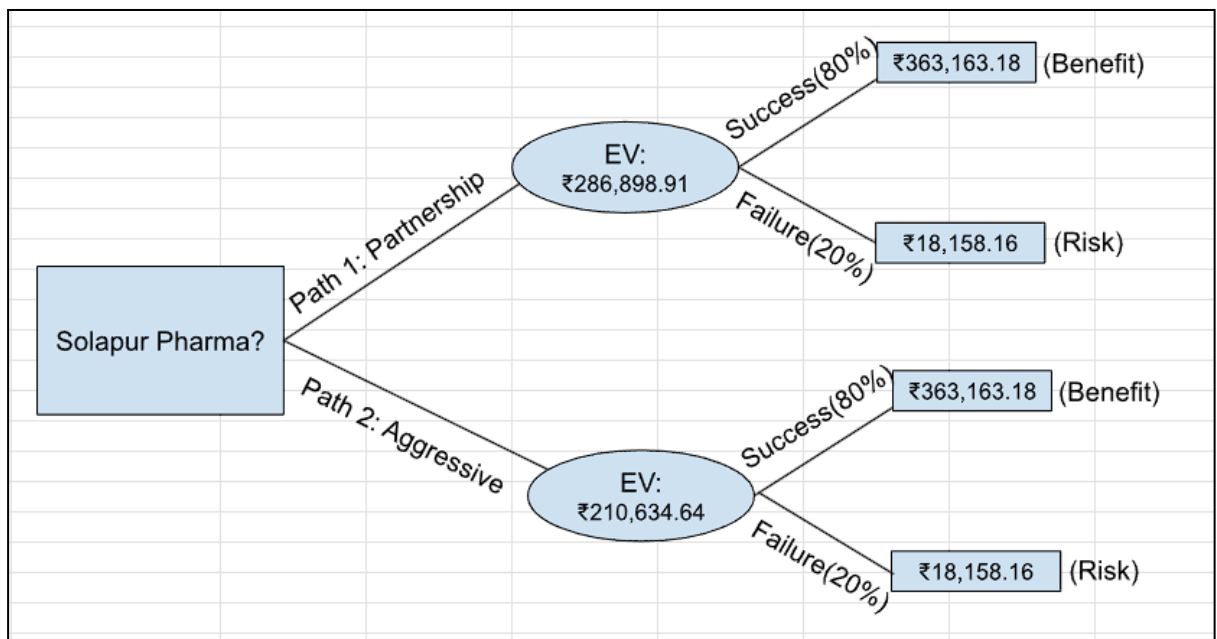


Figure 1: Solapur Pharma Decision Tree

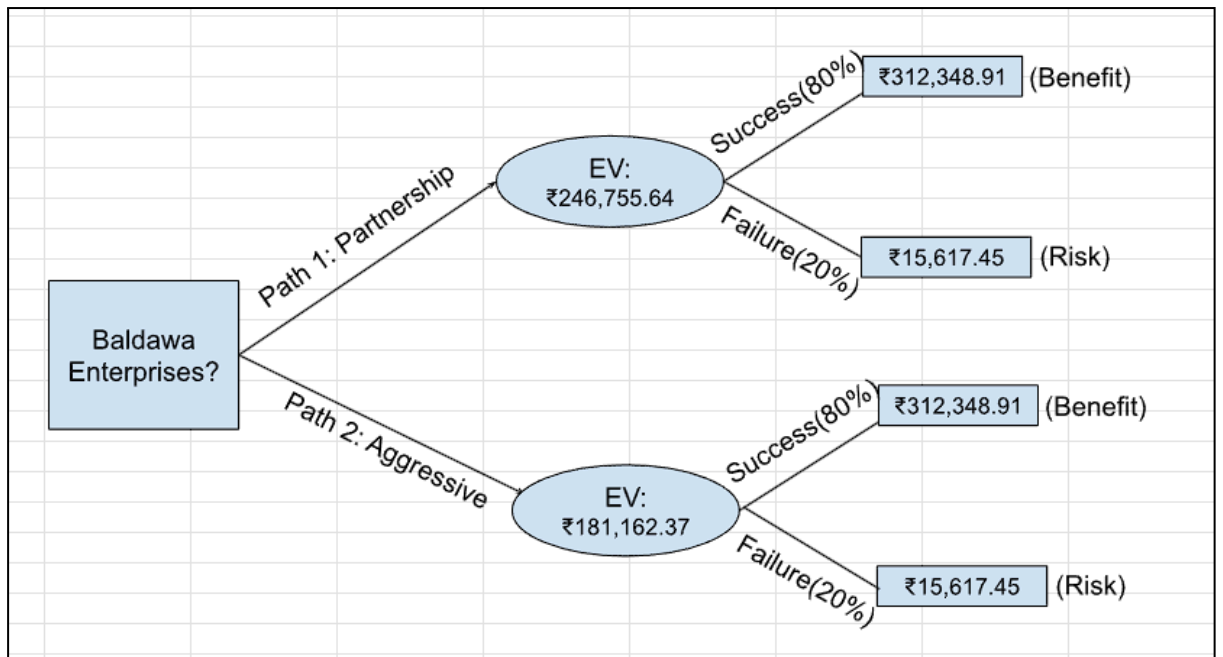


Figure 2: Baldawa Enterprises Decision Tree

Insights:

- The Decision Tree diagram (Figure 1) clearly shows that the "Partnership" strategy is the superior financial choice for both high-priority suppliers, as it yields a significantly higher Expected Value in both scenarios.
- As shown for Solapur Pharma LLB in Figure 1, the "Partnership" path's EV of ₹275,921 is over ₹87,000 higher than the "Aggressive" path's EV, clearly identifying it as the optimal choice.
- Similarly for Baldawa Enterprises (Figure 1), the "Partnership" path yields an EV of ₹235,478, which is far more favorable than the "Aggressive" path's EV of ₹158,609.
- This model demonstrates that the "Aggressive" strategy is a high risk, low reward option. Although Figure 1 shows it has a 60% chance of success, the 40% risk of a costly price hike severely reduces the potential return, making the "Partnership" path the clear data backed recommendation.

3.2. New Branch 12 Month Cash Flow & Sensitivity Analysis:

This analysis projects the financial performance of the new branch over its first 12 months, based on the ₹1,600,000 setup cost and ₹750,000 target annual profit. The model shows a stable Monthly Net Cash Flow of ₹62,500. However, due to the high initial setup cost, the Cumulative Cash Flow (Table 1) remains negative, ending the first year at -₹850,000, indicating the break-even point is not reached in Year 1. The sensitivity analysis (Figure 3) further reveals that a 10% drop in sales would result in an annual profit of only ₹612,500, while a 1% increase in COGS margin would drop the profit to ₹667,500.

TORNADO CHART DATA			
Variable	Downside Impact (-)	Upside Impact (+)	Base Annual Profit (Ref)
Sales (+/- 10%)	₹510,000.00	₹840,000.00	₹75,000.00
Fixed OpEx (+/- 10%)	₹585,000.00	₹765,000.00	₹75,000.00
COGS Margin (+/- 1%)	₹592,500.00	₹757,500.00	₹75,000.00

Table 1: New Branch 12-Month Cash Flow Projection

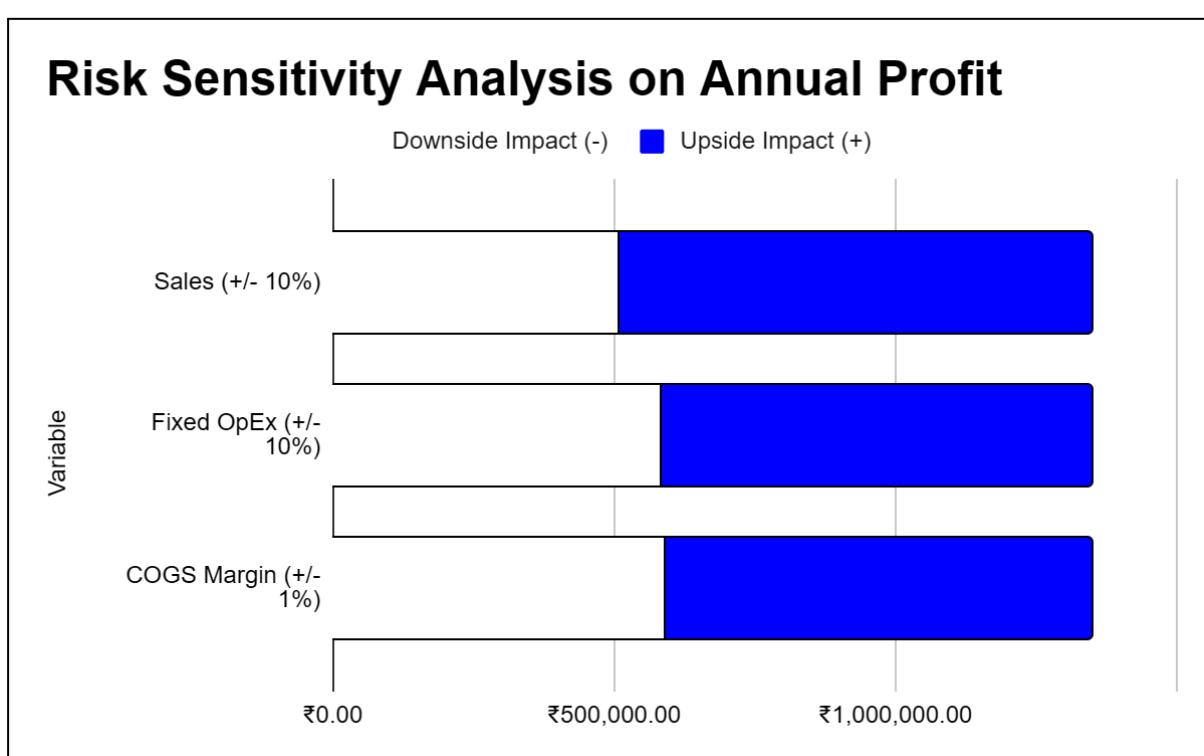


Figure 3: Risk Sensitivity Analysis on Annual Profit

Insights:

- The 12-Month Cash Flow model (Table 1) confirms that while the new branch is projected to be profitable on a monthly basis (₹62,500), the initial ₹1,600,000 setup cost will not be recouped in the first year.
- The Cumulative Cash Flow row in Table 1 is critical, as it shows the owner the exact cash position over time and demonstrates that the break-even point will be reached in the second year of operation, provided the projections hold.

- c. The Tornado Chart (Figure 3) provides a clear visual ranking of the project's financial risks. It identifies Sales as the single most impactful variable, where a 10% change can alter annual profit by ₹137,500.
- d. Figure 3 also shows that changes in COGS Margin (₹82,500 impact) and Fixed OpEx (₹90,000 impact) are significant, but secondary to the primary importance of maintaining sales volume. This clearly directs the owner's focus to sales and marketing as the number one priority for success.

3.3. Main Branch Profitability & Reinvestment Capacity Analysis:

This analysis was conducted to validate the financial stability of the main branch and its capacity to fund the expansion as per Scenario 1. The model tracks the Monthly Net Profit (bars) and the Cumulative Net Profit (solid line) over the 10-month period. The analysis shows the cumulative profit reached a total of ₹1,263,487.65 by January 2025, easily surpassing the ₹454,373 funding target required for the new branch.

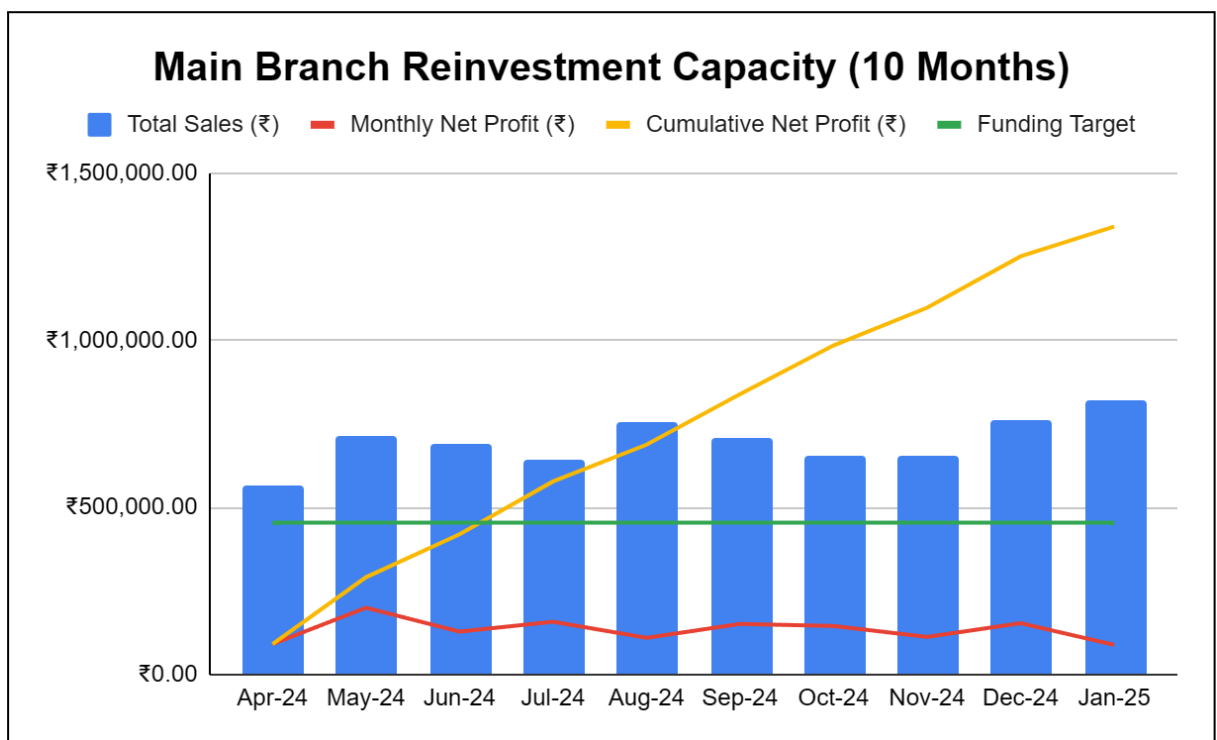


Figure 4: Main Branch Profitability and Reinvestment Capacity

Insights:

- a. Figure 4 provides a clear visual confirmation that the main branch is financially healthy and capable of funding its share of the expansion. The Cumulative Net Profit (solid line) rises steadily over the 10-month period.
- b. The chart clearly shows the Cumulative Net Profit surpasses the ₹454,373 Funding Target (red dashed line) by August 2024, only five months into the recorded period.

- c. By the end of January 2025, as shown in Figure 4, the main branch generated a total cumulative profit of ₹1,263,487.65. This is more than 2.7 times the amount required, demonstrating significant financial capacity and low risk for the reinvestment.
- d. The Monthly Net Profit bars in Figure 4 illustrate that the business is consistently profitable, which provides the stability needed to confidently re-allocate capital without endangering the primary operation.

For analysis/methods of Problem Statement 2:

3.4. Monthly Supply vs. Demand Imbalance Analysis:

This analysis provides a visual diagnosis of the inventory problem by plotting monthly Purchase Quantity (Supply) against Sale Quantity (Demand) for two key problem items. For LANTUS VIAL (Figure 5), the data shows supply consistently outpaced demand, with purchase spikes in September (40) and October (45) far exceeding sales. Conversely, for STARVOG 0.2mg (Figure 6), demand frequently exceeded monthly purchases, such as in April (35 sales vs 20 purchases) and September (28 sales vs 10 purchases), highlighting a clear stockout risk.

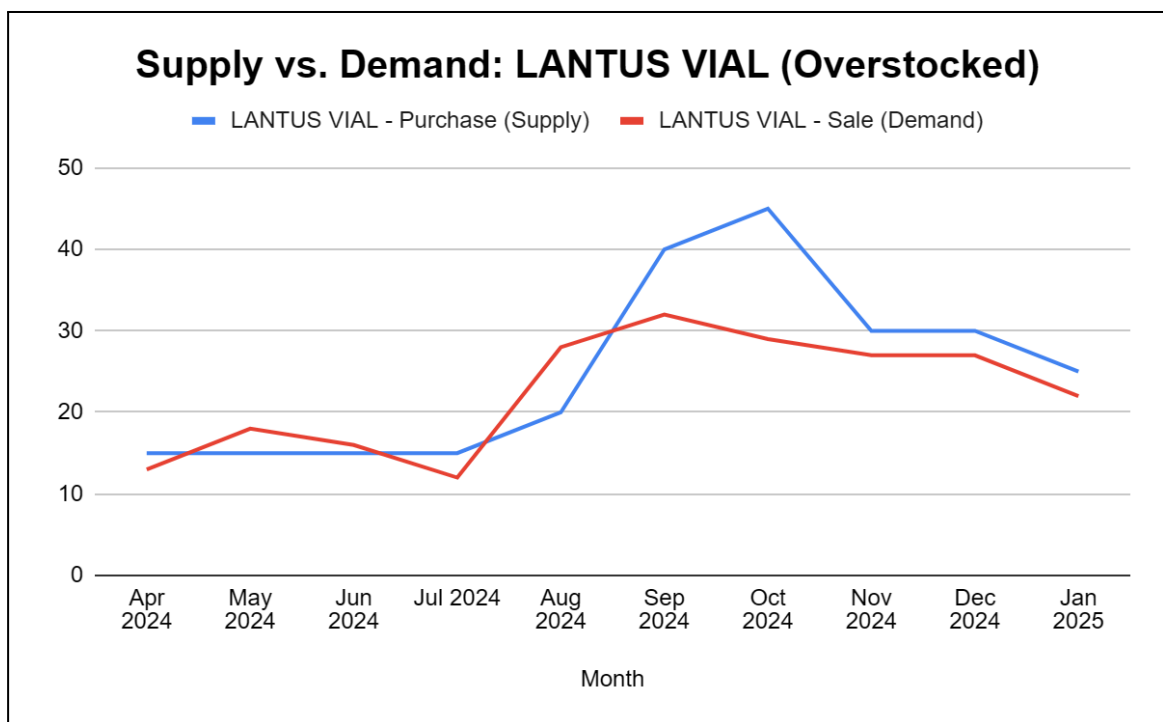


Figure 5: Supply vs. Demand: LANTUS VIAL (Overstocked)

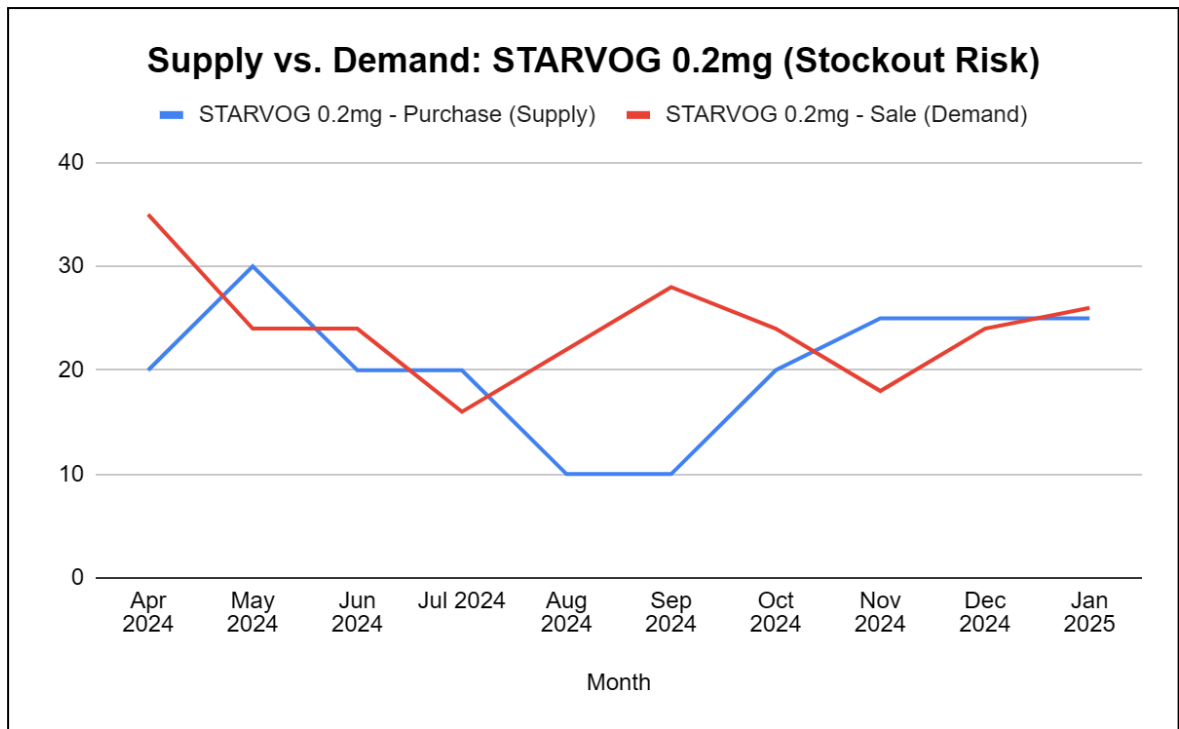


Figure 6: Supply vs. Demand: STARVOG 0.2mg (Stockout Risk)

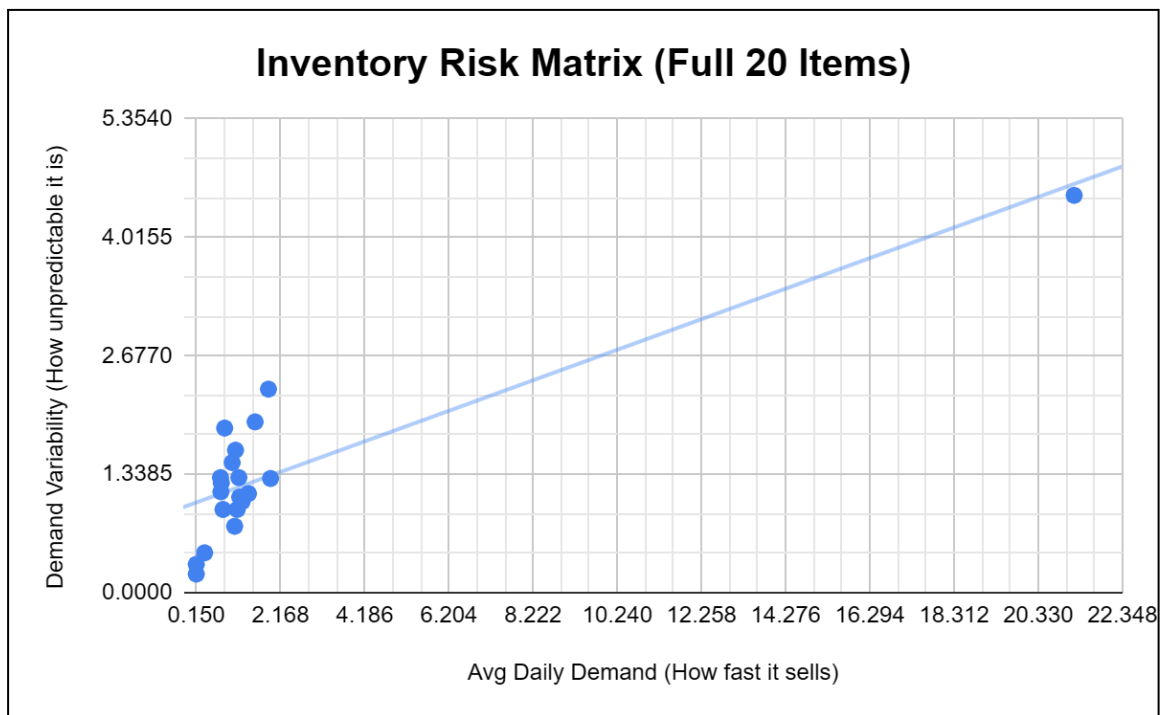
Insights:

- Figures 5 and 6 visually confirm the inventory imbalance problem identified in the midterm. They show the dynamic, month to month behavior that leads to the poor Inventory Turnover Ratio (ITR) findings.
- In Figure 5, the "Purchase (Supply)" line for LANTUS VIAL is consistently above the "Sale (Demand)" line, especially from September to January. This demonstrates a clear pattern of excessive purchasing that leads to overstocked inventory.
- In contrast, Figure 6 shows the "Sale (Demand)" line for STARVOG 0.2mg frequently spiking above its "Purchase (Supply)" line. This is evident in April, August, and September, illustrating a reactive purchasing model that fails to keep up with demand and creates a high risk of stockouts.
- Together, these two charts prove that the pharmacy's current purchasing method is based on guesswork, not data. It leads to simultaneous overstocking of some items and insufficient stock of others, which is inefficient and costly. This justifies the need for the statistical model in the next section.

3.5. Safety Stock (SS) & Reorder Point (ROP) Statistical Model:

This analysis created a statistical model to define a new, data driven inventory policy for all 20 medicines. The model calculated Average Daily Demand and Demand Variability to plot all items onto a risk matrix. The results, seen in Figure 7, immediately identified METSMALL 500mg (with an Avg Daily Demand of 21.19) as a massive outlier requiring its own management strategy. Figure 8 provides a "zoomed

in" view of the 19 other items, plotting them against the cluster's average demand (2.02) and variability (1.34).



Insights:

- The risk matrix charts (Figures 7 and 8) successfully categorize all 20 medicines into four distinct management quadrants, moving beyond a one size fits all inventory policy.
- Figure 7 clearly illustrates that METSMALL 500mg is a high demand outlier, with an average daily demand of 21.19. This is more than 10 times the cluster average of 2.02, proving it must be managed as a unique, high priority item.
- Figure 8 provides a clear, "zoomed in" view of the 19 'main cluster' items. It identifies the most problematic items, such as DGSON VM 500, DIVIFY MD, and JUSLINA 5mg, in the top right "High Demand/High Variability" quadrant, which require the most urgent management attention.
- The model in Figure 8 also effectively segments other items. It identifies stable sellers like FORTIUS (High Demand/Low Variability) and unpredictable, slow movers like GLUCORYL-M 0.5mg (Low Demand/High Variability), allowing for a specific and cost effective inventory policy for each.

3.6. Financial Impact of Inventory Optimization Analysis:

This final analysis quantifies the financial benefit of implementing the new Safety Stock model. The model calculates the Potential Savings by comparing the Current Capital Trapped on the shelf to the new Optimized Capital level. The analysis reveals that just the top 10 most overstocked items are trapping over ₹54,558 in unnecessary capital. METSMALL 500mg was the worst offender, with ₹18,556 in potential savings from that one item alone.

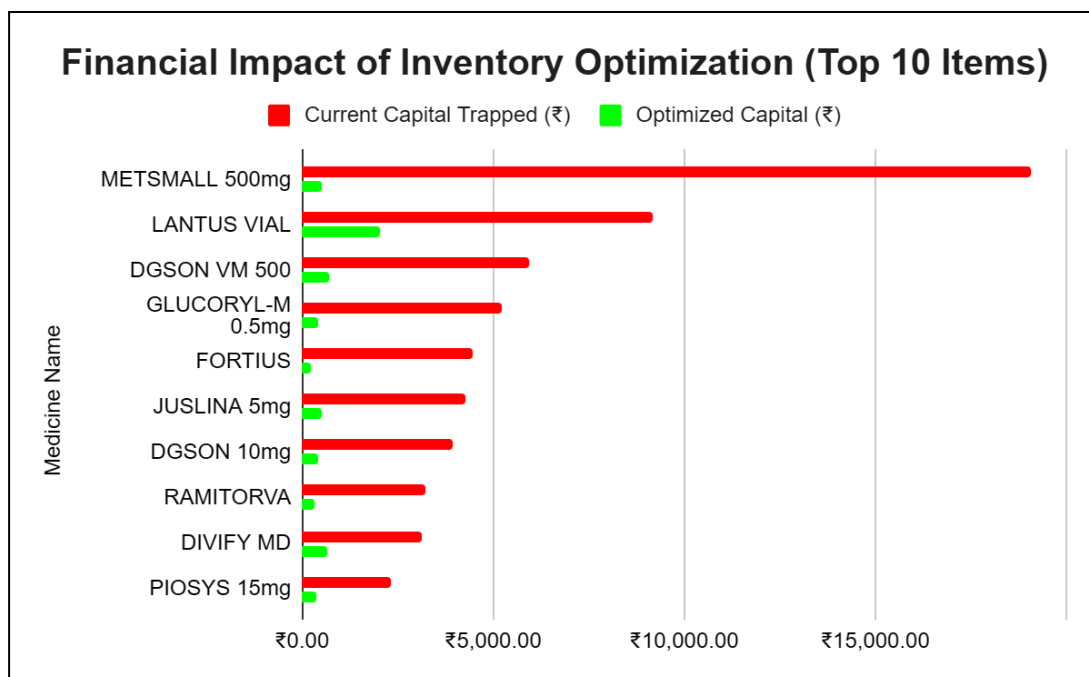


Figure 9: Financial Impact of Inventory Optimization (Top 10 Items)

Insights:

- a. Figure 9 provides a clear "before and after" visualization, comparing the Current Capital Trapped (blue/red bar) with the Optimized Capital (green bar). It clearly shows the significant reduction in cash tied up on the shelf for each of the top 10 items.
- b. The total potential savings from just these 10 items is ₹54,558. This is a substantial amount of liquid cash that is currently trapped in inefficient inventory and could be unlocked to help fund the new branch, directly linking the solution for Problem 2 back to Problem 1.
- c. The chart (Figure 9) shows that METSMALL 500mg is the single biggest opportunity. The pharmacy currently holds ₹19,070 of this item, but the new model shows it only needs ₹514, unlocking ₹18,556.
- d. This analysis also confirms our earlier findings. LANTUS VIAL and DGSON VM 500, identified as problematic in earlier analyses, are shown in Figure 9 to have potential savings of ₹7,139 and ₹5,221 respectively, proving the significant financial harm of the old inventory policy.

4. Interpretation of results and Recommendations

4.1. Interpretation and Recommendations for Problem Statement 1

The combined findings from Analyses 1, 2, and 3 provide a complete and financially sound solution to the pharmacy's primary scalability problem. The analysis proves that expansion is not only possible but can be achieved without external loans, thereby maximizing the return.

First, Analysis 3 (Main Branch Profitability) confirmed that the existing business is financially stable and highly profitable, generating over ₹1,260,000 in cumulative profit over 10 months. This result validates the first part of the "Scenario 1" funding plan: the main branch can easily and safely provide its ₹454,373 share of the capital.

Second, Analysis 1 (Decision Tree) solved the other half of the funding puzzle. It provided a clear, data backed strategy for unlocking the ₹1,145,626 in trapped supplier capital. By modeling the financial outcomes, it proved that a low conflict "Partnership" strategy is the optimal path, yielding a high Expected Value of ₹275,921 from Solapur Pharma and ₹235,478 from Baldawa Enterprises.

Finally, Analysis 2 (Cash Flow & Sensitivity) provided a realistic, forward looking model of the new branch's performance. It confirms that the branch will

be profitable on a monthly basis (₹62,500) but will not break even in its first year, setting a clear long term expectation. Critically, the Tornado Chart identified "Sales" as the single most important variable for success, providing a clear focus for all future operational efforts.

Recommendations: Based on this interpretation, the following recommendations are proposed to solve Problem Statement 1:

- **Immediately Initiate "Partnership" Negotiations:**

Specific: The owner should personally initiate strategic "Partnership" discussions with the leadership at Solapur Pharma LLB and Baldawa Enterprises.

Measurable: The goal is the successful extension of payment terms to unlock the identified ₹1,145,626 in trapped capital.

Achievable: This is highly achievable. The "Partnership" path has a high (80%) assumed success rate and a high Expected Value, making it a favorable proposition for both parties.

Relevant: This action is the primary enabler of the entire self-funded expansion plan, which provides the highest ROI.

Time bound: These negotiations should be initiated within the next 30 days.

- **Earmark Main Branch Profits for Expansion:**

Specific: The owner should formally earmark ₹454,373 from the main branch's cumulative profits to be used as the second component of the new branch's setup cost.

Measurable: The successful allocation and "ring fencing" of these funds.

Achievable: This is 100% achievable, as Analysis 3 showed the main branch has over 2.7 times this amount in available profit.

Relevant: This is the final step to fully fund Scenario 1 without incurring any loan interest.

Time bound: This capital should be made liquid and ready for transfer as soon as the expansion is green lit.

- **Establish "Sales" as the New Branch's Key Performance Indicator (KPI):**

Specific: All operational focus for the new branch must be on achieving the ₹687,500 "Base Monthly Sales" target identified in Analysis 2.

Measurable: The monthly sales figures must be tracked against this specific target.

Achievable: This target is the baseline required to hit the ₹750,000 annual profit goal, making it a realistic and necessary target.

Relevant: The Sensitivity Analysis (Tornado Chart) proved that "Sales" is the single most critical factor determining the new branch's profitability.

Time bound: This KPI must be the primary focus from Day 1 of the new branch's operation and be reviewed monthly for the first 12 months.

Implementation and Benefits: Implementing these recommendations provides a clear, three step playbook for growth. The first two steps secure the full ₹1,600,000 in setup costs before any commitment is made, completely de-risking the expansion from a funding perspective. The third step provides a clear operational focus to manage risk after launch. The primary benefit is that the pharmacy can confidently solve its scalability problem and open a new branch by following a plan that is entirely self funded, avoids interest payments, and maximizes the potential return on investment.

4.2. Interpretation and Recommendations for Problem Statement 2

The findings from Analyses 4, 5, and 6 provide a comprehensive diagnosis and a clear, data driven solution to the pharmacy's chronic inventory mismanagement problem. The analyses prove that the pharmacy is trapping significant capital in overstocked items while simultaneously risking lost sales from stockouts of critical medicines.

First, Analysis 4 (Supply vs. Demand) provided a powerful visual confirmation of the problem. By tracking LANTUS VIAL and STARVOG 0.2mg, it showed the real world consequence of the pharmacy's guesswork purchasing model, which leads to supply and demand being constantly mismatched. This set the foundation for why a new system is necessary.

Second, Analysis 5 (Safety Stock & ROP Model) provided the technical solution. By calculating the Average Daily Demand and Demand Variability for all 20 medicines, it created a statistical model that replaces guesswork with an objective, data driven policy. The Risk Matrix charts (Figures 7 and 8) successfully categorized the entire inventory, identifying high-risk items (like

DGSON VM 500) and massive outliers (like METSMALL 500mg) that require special management.

Finally, Analysis 6 (Financial Impact) provided the critical business case for making these changes. It quantified the financial harm of the current system by showing that over ₹54,558 in cash is trapped in just the top 10 overstocked items. It proved that adopting the new model is not just an operational improvement but a significant financial opportunity, directly linking the solution for Problem 2 back to the goals of Problem 1.

Recommendations: Based on this interpretation, the following recommendations are proposed to solve Problem Statement 2:

- **Immediately Adopt the Safety Stock & Reorder Point (SS & ROP) Model:**

Specific: Replace the current "guesswork" purchasing system with the new policy. This means placing a new order for an item only when its stock level hits the Calculated Reorder Point (from Analysis 5).

Measurable: The adherence to the new ROP numbers for all 20 medicines. A secondary metric will be the reduction in both inventory value and stockout incidents over the next 6 months.

Achievable: This is highly achievable. The model provides the exact, objective numbers for all 20 medicines, removing all guesswork from the process.

Relevant: This directly solves the core problem of stock imbalances by implementing a proven, statistical inventory management system.

Time bound: This new policy should be implemented for all new orders starting immediately.

- **Prioritize Unlocking Capital from Top 10 Overstocked Items:**

Specific: Place a temporary freeze on purchasing the "Top 10" overstocked items identified in Analysis 6 (Figure 9), including METSMALL 500mg, LANTUS VIAL, and DGSON VM 500.

Measurable: The goal is to unlock the identified ₹54,558 in trapped capital by allowing sales to reduce the current inventory down to the new "Optimized Capital" level.

Achievable: This is achieved by simply not re-ordering until the stock levels fall to the new Calculated Reorder Point.

Relevant: This action provides the most significant and immediate financial benefit, directly supporting the pharmacy's primary goal of improving its cash position for expansion.

Time bound: The purchasing freeze for these 10 items should be in effect immediately and last until their stock levels are normalized, which should be reviewed monthly.

Implementation and Benefits: Implementing these recommendations will fundamentally solve the pharmacy's inventory inefficiencies. The first recommendation creates a stable, efficient, and long term system for managing all items. The second recommendation provides an immediate, short term action plan to recover tens of thousands of Rupees in cash that is currently wasted. The combined benefits are significant: a drastic reduction in overstocking waste and expiry, fewer lost sales from stockouts, improved customer satisfaction, and, most importantly, a direct injection of newly unlocked capital that can be used to fund the pharmacy's expansion.